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DETAIL SPECIFICATION

VALVES, CHECK – BRONZE- AND STEEL-BODY

This specification is approved for use by all Departments and Agencies of Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bronze- and steel-body check valves with regrinding, renewable, or regrinding-renewable disks and seats.

1.2 Classification. The check valves are of the following groups, types, styles, trims, end connections, sizes, and classes suitable or not suitable for potable water use, as specified (see 6.2).

1.2.1 Groups.

Group	PIN Code	Description
A	A	Bronze-body valves (⅛ through 4 inches, see table I)
B	(none)	No longer applicable
C	C	Steel-body valves (¼ through 12 inches, see table II)

1.2.2 Types.

Type	PIN Code	Description
I	1-	Lift check, horizontal installation
IS	1S	Lift check, spring-loaded, horizontal installation
II	2-	Lift check, vertical installation
IIS	2S	Lift check, spring-loaded, vertical installation
III	3-	Swing check
IV	4-	Tilting-disk check
V	5-	Butterfly, spring-loaded, dual-disk check

1.2.3 Styles.

Style	PIN Code	Description
A	A	Disk type, swing- or post-guided flow control device (types I, IS, II, IIS, and III)
B	B	Piston type, cylinder-guided, air-cushioned flow control device (types I and II)
C	C	Ball type flow control device (types I and II)
N/A	-	As defined by the type (types IV and V)

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AMSC N/A FSC 4820

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1.2.4 Trims. (see table III)

Trim	PIN Code	Description
1	1-	Non-metallic disk; integral seat (group A)
2	(none)	No longer applicable
3	3-	Metallic, regrinding disk; integral seat (group A)
3A	3A	Metallic, nonregrinding disk; renewable (group A, type III)
4	4-	Metallic, regrinding disk; renewable seat ring (group A, type I)
5	5-	Metallic flow control device; renewable seat ring (group C)
6	6-	Metallic flow control device; integral or welded in seat (style C)
7	7-	As specified (wafer-body valves)

1.2.5 End connections. (see table IV)

End Connection	PIN Code	Description
T	T-	Threaded
F	F-	Flanged
S	S-	Solder (group A)
W	W-	Wafer-body (see 3.6.5)
BW	BW	Butt welding (group C)
SW	SW	Socket welding (group C, 2-inch and smaller)

1.2.6 Sizes. Check valve sizes are ⅛ inch through 24 inches (see tables I and II).

Size (nps)	PIN Code
⅛	001
¼	002
½	005
1	010
2	020
3	030
4	040
5	050
6	060
8	080
10	100
12	120

1.2.7 Classes. Check valve classes are 125 to class 300 (see tables I and II).

Class	PIN Code	Description
125	012	125 psi
150	015	150 psi
200	020	200 psi
300	030	300 psi

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1.2.8 Potable water use. (see 3.2)

PIN Code	Description
PW	For potable water use
(blank)	Not for potable water use

1.3 Part or identifying number (PIN). The PINs for check valves are created as follows:

M	18436	-	X	XX	X	XX	XX	XXX	/	XXX	XX
Prefix for Military Specification	Specification Number	-	Group (see 1.2.1)	Type (see 1.2.2)	Style (see 1.2.3)	Trim (see 1.2.4)	End Connection (see 1.2.5)	Size (see 1.2.6)	/	Class (see 1.2.7)	Potable Water (see 1.2.8)
Example: M18436-A1-B3-F-030/012PW is for a bronze lift check, spring-loaded, horizontal, piston type flow control, metallic regrinding disc, flanged end, 3-inch, class 125 valve suitable for potable water use.											

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-PRF-20042 - Flanges, Pipe, Bronze (Silver Brazing)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-798 - Nondestructive Testing, Welding, Quality Control, Material Control and Identification, and Hi-Shock Test Requirements for Piping System Components for Naval Shipboard Use

(Copies of these documents are available online at http://quicksearch.dla.mil.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AQ-GIB-010/248	-	Requirements for Welding and Brazing Procedure and Performance Qualification
S9074-AR-GIB-010/278	-	Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/ DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME Y14.36M	-	Surface Texture Symbols
ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of these documents are available online at http://webstore.ansi.org, www.asme.org, or http://asq.org/quality-press/index.html.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.20.1	-	Pipe Threads, General Purpose, Inch
ASME B16.5	-	Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard
ASME B16.10	-	Face-to-Face and End-to-End Dimensions of Ferrous Valves
ASME B16.18	-	Cast Copper Alloy Solder-Joint Pressure Fittings
ASME B16.24	-	Bronze Pipe Flanges and Flanged Fittings
ASME B16.25	-	Buttwelding Ends
ASME B16.34	-	Valves Flanged, Threaded and Welding End
ASME B31.1	-	Power Piping

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A182/A182M	-	Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service
ASTM A276/A276M	-	Standard Specification for Stainless Steel Bars and Shapes
ASTM A351/A351M	-	Standard Specification for Castings, Austenitic, for Pressure-Containing Parts

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ASTM A479/A479M	-	Standard Specification for Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels
ASTM A582/A582M	-	Standard Specification for Free-Machining Stainless Steel Bars
ASTM B61	-	Standard Specification for Steam or Valve Bronze Castings
ASTM B62	-	Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM B140/B140M	-	Standard Specification for Copper-Zinc-Lead (Red Brass or Hardware Bronze) Rod, Bar, and Shapes
ASTM B164	-	Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire
ASTM B584	-	Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM D2240	-	Standard Test Method for Rubber Property—Durometer Hardness

(Copies of these documents are available online at www.astm.org.)

MANUFACTURERS STANDARDIZATION SOCIETY (MSS) OF THE VALVE AND FITTINGS INDUSTRY

MSS SP-25	-	Standard Marking System for Valves, Fittings, Flanges, and Unions
MSS SP-42	-	Corrosion-Resistant Gate, Globe, Angle, and Check Valves with Flanged and Butt Weld Ends (classes 150, 300, & 600)
MSS SP-45	-	Bypass and Drain Connections
MSS SP-61	-	Pressure Testing of Valves
MSS SP-80	-	Bronze Gate, Globe, Angle, and Check Valves
MSS SP-126	-	In-Line, Spring-Assisted, Center-Guided Check Valves (Carbon, Alloy Steel, Stainless Steel, & Nickel Alloys)

(Copies of these documents are available online at http://www.msshq.org.)

NSF INTERNATIONAL

NSF/ANSI 61	-	Drinking Water System Components - Health Effects

(Copies of this document are available online at https://www.techstreet.com/nsf/.)

UNDERWRITERS LABORATORES, INC. (UL)

UL 312	-	Standard for Check Valves for Fire-Protection Service

(Copies of this document are available online at www.ul.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2. New design valves, or redesign of existing valves, shall be subjected to first article inspection.

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3.2 Materials. Materials shall be specified as follows. Materials used shall be free from defects that would adversely affect the performance or maintainability of individual components or of the overall assembly. NSF/ANSI 61 has stringent lead limitations on surfaces in contact with drinking water. Actual materials used for check valves to be employed in drinking (potable) water systems shall meet the definition of "lead free" in accordance with NSF/ANSI 61 by not containing more than 0.2 percent lead when used with respect to solder and flux and not more than a weighted average of 0.25 percent lead when used with respect to the wetted surfaces of pipes, pipe fittings, plumbing fittings, and fixtures.

3.2.1 Bronze. Bronze for group A valve bodies and caps shall be in accordance with MSS SP-80.

3.2.2 Steel. Steel for group C valves shall be selected from steel material in accordance with ASME B16.34.

3.2.3 Copper-nickel alloy. Copper-nickel alloy specified in table III for disks and seat rings shall be in accordance with ASTM B584, alloy No. C97600 to C97800. Alternate copper-nickel alloys used commercially for valve trim will be acceptable provided alloying elements are in the range of 57 to 63 percent for copper and 24 to 35 percent for nickel.

3.2.4 Nickel-copper alloy. Nickel-copper alloy specified in table III as an optional trim material shall be in accordance with ASTM B164, class B.

3.2.5 Stainless steel. When specified (see 6.2), stainless steel trim specified in table III for disk and seat rings on steel-body valves shall be type 304, 316, CF3M, or CF8M in accordance with ASTM A182/A182M, A276/A276M, A351/A351M, or A582/A582M, as applicable. Application of stainless steel facing by welding is permitted where 13 percent chrome filler metal is used. Stainless steel parts shall be type 410, 416, 420, or 440 in accordance with ASTM A276/A276M, ASTM A479/A479M, or ASTM A582/A582M, as applicable.

3.2.6 Spring material. Springs furnished on type IS and IIS spring-loaded lift check valves and on type IV and V valves shall be stainless steel, nickel-chromium, or nickel-copper alloy except that on bronze-body valves (group A), phosphor bronze springs shall be acceptable.

3.2.7 Wrought copper alloys. Wrought copper alloys specified herein as alternate materials for caps, disks, disk holders, and disk hangers on valves less than ¾-inch nominal size shall be in accordance with ASTM B140/B140M, UNS alloy C31400 or C31600. Wrought copper alloys shall not be used as an alternate to ASTM B61 for pressure-containing parts on valves having a service temperature rating exceeding 400 °F.

3.3 Operational characteristics. The check valves shall consist essentially of a body, flow control device, and, except as otherwise specified herein, a removable threaded cap or bolted cover. The valves shall permit flow in one direction only and shall close automatically when flow is reversed. These valves shall be of the lift, swing, tilting, or dual-disc construction. The valve seats, except for trims 6 and 7, shall be either regrinding or renewable. The inter-relationship of groups, types, styles, trims, end connections, sizes, and classes shall be as specified in 1.2 and tables I, II, III, and IV.

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TABLE I. Group A, bronze-body check valves.

End connections			Steam rating [1/]		WOG rating (psig) [4/]
Class	Type [2/]	Size (inches)	psig	°F [3/]	
125	T, S [5/]	⅛-3	125	353	200
150	T, S [5/]	⅛-4	150	366	300
150	F	⅛-4	150	366	225
200	T	⅛-3	200	550	400
	T	⅛-3	300	550	600
300	F	1-2	300	550	500

NOTES:

[1/] Steam rating: The steam rating consists of two mutually interdependent conditions: pressure and temperature. The specified non-shock steam pressure rating in pounds per square inch gauge (psig) is applicable when the coincident temperature conditions are expressed in conjunction with the pressure rating to give the overall steam rating. In the actual application of bronze and steel-body valves, the pressure can be increased up to the water, oil, gas (WOG) rating as the temperature decreases. In no case, except for steel-body valves, can the temperature be increased above the maximum steam temperature rating specified herein. The class designations used herein reflect the steam pressure ratings and are equivalent to ratings expressed in terms of maximum operating pressure or maximum working pressure. The class designations applied herein to steel-body valves are referred to in ASME B16.34 as the primary service pressure ratings.

[2/] Legend for types of end connections: T-Threaded, F-Flanged, S-Solder.

[3/] Non-metallic disks, if specified, may reduce maximum temperature rating (see tables III and IV).

[4/] WOG rating: The WOG rating is the maximum non-shock pressure, expressed in psig for which the pressure-containing parts of the valve are rated when the coincident media temperature is between -20 °F and 150 °F for bronze-body and -20 °F and 100 °F for steel-body valves. The WOG rating indicates the ability of the valve to withstand the pressure effect of water, oils, and gases and does not imply that a WOG-rated valve will necessarily withstand the potentially deleterious effect of all aqueous solutions, petroleum products, and gaseous media unless appropriate material selectivity has been exercised.

[5/] The safe pressure: Temperature rating of a solder-joint pipe system is dependent not only on valve, fitting, and tubing strength, but also on the composition of the solder used. It shall be the responsibility of the user to select a solder composition that is compatible with service conditions.

TABLE II. Group C, steel-body check valves.

End connections			Steam rating [1/]		WOG rating (psig) [4/]
Class	Type [2/]	Size (inches) [3/]	psig	°F [4/]	
150	T	¼-4	150	500	275
150	F, BW	¼-12	150	500	275
300	T	¼-3	300	850	720
300	F, BW	¼-12	300	850	720

NOTES:

[1/] Steam rating: See table I, note 1.

[2/] Legend for types of end connections: T-Threaded, F-Flanged, BW-Butt welding.

[3/] ¼-inch up to 2-inch are socket welding.

[4/] WOG rating: See table I, note 4.

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TABLE III. Trim requirements.

Group	Valve applicability				Material requirements for disks & seat rings [1/]	
	Trim	Type	Style	Classes	Standard	Optional (see 6.2)
A (Bronze-body)	1	I	A	150, 200	ND1 [2/]	None
		IS	A	150, 200	ND1	None
		II	A	150	ND1	None
		IIS	A	150	ND1	None
		III	N/A	125, 150, 200	ND1	None
	3	I	A	150, 200, 300	Bronze disk	None
		I	B	300	Bronze disk	None
		II	A	125, 200	Bronze disk	None
		III	---	125, 150, 200, 300	Bronze disk	None
	4	I	A	150, 200, 300	Copper-nickel alloy disk and seat ring	Stainless steel or nickel-copper alloy disk and seat ring
	6	I	C	125, 150, 200	Stainless steel or bronze ball	None
		II	A	125	Bronze disk	None
		III	C	125, 200	Stainless steel or bronze ball	None
C (Steel)	N/A	III	---	150, 300	Stainless steel disk and stainless steel, threaded seat rings. Hard faced steel seat rings shall be seal welded in the body.	Nickel-copper alloy, bronze or hard faced disks and seats; seat shall be hard faced weld-deposited on body.

NOTES:

1/ Trims 1 and 3 include integrally cast, bronze seats.

2/ Non-metallic discs shall be Tetrafluorethylene (Teflon, Fluon, or equal PTFE), 400 psig, and 400 °F. Non-metallic Disc (ND) applications include oxygen; acids, alkalis, and other corrosive solvents; gasoline; oil; gas; steam; and hot and cold water.

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TABLE IV. End connections.

End connection	Group (see 1.2.1)	Classes	Applicable ASME standard	Flange facing [1/]
Threaded	A	All	B1.20.1	---
	C	150, 300	B1.20.1	---
Flanged [2/]	A	150, 300	B16.24	Plain face [3/]
	C	150, 300	B16.5	1/16-inch raised face
Solder	A	125,150	B16.18	---
Butt welding	C	All	B16.25	---

NOTES:

1/ Standard facings; alternate facings to be furnished only when specified (see 6.2).

2/ When specified (see 3.6.5 and 6.2), flange drilling patterns in accordance with MIL-PRF-20042 shall be used on bronze check valves.

3/ Plain face flanges shall be smooth-faced without projection or raised rings.

3.3.1 Lift checks. Lift checks are of type I, horizontal installation; type IS, spring loaded horizontal installation; type II, vertical installation; and type IIS, spring-loaded vertical installation. The design of lift check valves shall be in conformance with the configurations illustrated in MSS SP-80 or MSS SP-126, depending on valve type and material. The seating surface in type I, IS, II, and IIS lift check valves shall be located on the horizontal segment of an integral bridge wall separating the lower inlet chamber from the upper outlet chamber so that incoming flow will lift the flow control device upward from the seat. The flow control device shall be a beveled or flat metallic disk; a non-metallic disk secured in a metallic disk holder; a metallic, piston-shaped or piston-guided disk; or a spherical metal ball in accordance with the specified style and trim. On style B valves, the design of the piston disk or guide shall provide a cushioning effect to minimize hammering resulting from repeated openings encountered under pulsating flow conditions. Spring-loaded type IS and IIS valves shall also be designed to cushion the disk on opening. The spring action on these valves shall, in addition, serve to affect rapid closure to eliminate hydraulic hammer of the disk caused by frequent flow reversals. On all lift check valves, the lift of the flow control device in the direction perpendicular to the seat shall be no less than ¼ the diameter of the seat port opening.

3.3.2 Swing checks. Swing checks are type III check valves. The closure assembly on swing check valves shall consist essentially of a strap or hanger, hinge pins, disk bolt and nut, and a metallic disk with a non-metallic facing or encapsulated steel disk with a flexible hinge portion that bends a flexer as the flow opens up the valve. In type III valves, the fluid flow shall swing the closure assembly away from the seat. The seating mechanism on all swing check valves shall be such that the disk assembly will close by the force of gravity when the valve is installed in a horizontal line. On regrinding type valves the seats shall be inclined such that the angle between the central axis perpendicular to the seating surface and the centerline of the stream flow is not less than 30 degrees or more than 60 degrees. All type III valves, except those with wafer bodies, shall be equipped with stops on the body, cover, or hanger strap to prevent disks from sticking or wedging in the wide open position. Swing check valves shall be applicable for installation both in a horizontal run of piping and for vertical runs in which the flow is the upward direction. Valves having a clear-way swing check shall have face-to-face and end-to-end dimensions in accordance with UL 312. Swing check valves shall be designed in accordance with MSS SP-80 (bronze). Corrosion-resistant check valves with flanged and butt weld ends shall be designed in accordance with MSS SP-42.

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3.3.3 Tilting-disk checks. Tilting-disc checks are type IV check valves. When specified (see 6.2), the tilting-disk check valve shall be a free-acting, cast-split body design or a spring-loaded cylindrical body design. The closure assembly on type IV, tilting disk check valves shall have a bottom hydraulic snubber designed to contact the disk during the last 10 percent of closure and control the final closing of the valve to prevent water hammer. The pivot point for the disk shall be located between the centerline of the flow and the top of the disk, and shall be offset from the plane of the seating surface. The housing shall be equipped with flanges or welding ends, or shall be of wafer-type construction in accordance with the end connections specified.

3.3.4 Dual-disc checks. Dual-disc checks are type V check valves. The dual-disc check valve mechanism shall consist of two half-circle disks hinged together. The two disks fold together upon positive flow and retract to a full circle to close against reverse flow resisted by a spring that is held in place by a retainer. As upstream pressure increases and overcomes the spring force, the discs fold off their seats allowing flow through the valve. The spring allows the valve to be installed in any direction. These should not be confused with a double check valve, which is an assembly that contains two distinct check valves.

3.4 Identical items. All valves of the same classification shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. Disks and seats shall be lapped to effect satisfactory replaceability of renewable seating surfaces.

3.5 Design and performance. All steam-rated check valves shall be designed for the applicable steam pressure and coincident temperature specified herein.

3.6 Detail of components.

3.6.1 Bodies. Bodies for bronze and steel valves shall be a unidirectional flow valve having a single disk and a seat that provide a controlled closing rate, proportional to the flow. Vertical, angle pattern valves shall be amply proportioned with the vertical axis of the cap parallel to the stream flow in which the inlet and outlet are perpendicularly right-angle. Horizontal, bronze-body swing check valves shall be of the conventional T-pattern with the vertical axis of the cap inclined at an angle of 30 to 50 degrees with the stream flow. Bronze-body, T-pattern, regrinding valves shall be equipped with a threaded, plugged, regrinding port. The minimum wall thickness of steel valves of threaded, flanged, and butt welding construction shall be in accordance with ASME B16.34. For flanged, cast valves, the flanges shall be cast integrally with the body. For flanged, forged steel valves, the flanges shall be forged integrally with the body or the flanges may be forged separately and welded to the body. Welding to attach a forged flange shall be in accordance with S9074-AR-GIB-010/278 applicable to the welding of pipe joints.

3.6.2 Caps or cover plates. All type I, IS, and III check valves shall be equipped with a readily removable and replaceable threaded cap or bolted cover plate designed to permit access to and removal of replaceable working parts for the valve without removal of the valve from the line. Caps for bronze-body valves in sizes up to and including 3 inches, shall be of the inside screw, one-piece screw-over, union-ring, or bolted plate type. For all other valves, bolted cover plates shall be furnished. Caps and cover plates shall be fabricated of the same material or material superior to that specified herein for the body, except steel plate may be used on steel valves in lieu of the casting material specified herein for the bodies. Wrought copper alloys in accordance with 3.2.7 may be used on class 125 and 150 bronze-body valves ¾ inch and smaller in lieu of ASTM B62. Type II and IIS vertical lift check valves shall be designed with a separable, two-section, threaded-connected casing or shall otherwise be designed so that, after the valve is removed from the line, the ball or disk assembly can be removed. On bronze-body, Y-pattern valves with trim 3 or 4, the threaded cap shall provide access for regrinding operations. Caps or cover plates will not be required on type III valves furnished with wafer bodies.

3.6.3 Seats. Seats shall be integral or renewable in accordance with the trim specified. Material for seats shall be as specified in table III. Renewable seat rings shall be equipped with lugs, grooves, or similar means for removal and shall have sufficient threaded surface to prevent leakage between the seat ring and body under the applicable service conditions.

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a. Regrinding valve. On a regrinding valve (lapping), the seat can be reground using the metallic disk as the rotating surface to which the grinding compound is applied. For regrinding purposes, the stem on regrinding-type disks shall be slotted to receive a screw driver or, for style B valves, the disk shall be equipped with an internally threaded cylindrical section to receive a stem that will extend outside the valve for manual rotation. Except for vertical lift check valves, all hard faced valve seats and disks (trim 3, 4, and 5) shall be capable of being reground without removal of the valve from the pipeline. The regrinding feature does not apply to wafer-body swing checks, tilting disk checks, and butterfly checks.

b. Renewable valve. The term renewable valve, as applied herein, shall mean that valve parts known as disks and seat rings shall be affixed or contained by a threaded element and be readily removed for replacement. Seat rings requiring the use of special seat ring wrenches for removal shall be considered renewable. Pressed-in seat rings on wafer-body swing check valves that can be readily dislodged from the body casting by tapping shall be considered renewable. Seat rings shall be renewable when trims 4 and 5 are specified; all flow control devices, whether disks, pistons, or balls shall be renewable. Weld deposited seats that are regrindable will be acceptable on group C valves.

3.6.4 Flow control devices. The flow control device shall be a ball, piston, swing disk, lift disk, tilting disk, or butterfly disk in accordance with the specified valve type and, when applicable, style. Materials for the flow control devices shall be as specified in table III. All flow control devices shall be renewable. Disk hangers and disk holders on bronze-body valves shall be fabricated of bronze in accordance with ASTM B61 or B62. The disk holder for steel, except that for class 200 bronze-body valves, shall be in accordance with ASTM B61 only. Disk hangers and disk holders for valves ¾-inch or less with a service temperature rating not exceeding 400 °F may be fabricated of a wrought copper alloy specified in 3.2.7 in lieu of ASTM B61 or B62. All disks on style A lift check valves shall be guided both above and below the seat by wing guides, a single upper or lower post guide, or a combination of the two; except on 3-inch and smaller valves, a single upper or lower post guide will be acceptable.

3.6.5 End connections. End connections for type I through type IV check valves shall be as specified in table IV. Valves furnished with wafer-type bodies shall be applicable for installation between the standard flanges specified (see 6.2). Bolting for installation of wafer-body valves shall be located on the outside circumference of the valve and not through the peripheral area of the body itself as in conventional flanged connections. Face-to-face and end-to-end dimensions for flanged and butt weld classes of steel-body valves shall be in accordance with ASME B16.34. The end-to-end dimensions for threaded and socket welded steel-body valves shall be the manufacturer's standard. When specified (see 6.2), flange drilling patterns in accordance with MIL-PRF-20042 shall be used on bronze check valves.

3.7 Special features.

3.7.1 Bosses. When specified (see 6.2), flanged steel-body check valves 2 inches and larger shall be furnished with tapped bosses. The number and location of the bosses shall be in accordance with MSS SP-45 with respect to locations and designations. Tappings shall be of the size recommended in MSS SP-45 unless an alternate size is specified (see 6.2). Valves specified for shipboard use shall have welded bosses in accordance with S9074-AR-GIB-010/278 (see 6.2).

3.7.2 Operating levers. When specified (see 6.2), type III swing check steel-body valves over 2 inches in nominal size shall be equipped with an auxiliary operating lever. The lever shall be weight- or spring-loaded as specified (see 6.2). In either case, the torque on the lever shall be controlled either by manual movement of the weight or adjustment of the spring tension. The lever assembly shall be designed and initially installed to provide more rapid closure on flow reversal to reduce hydraulic hammering of the disk against the seat. The lever and weight assembly shall also be designed to be repositioned so as to provide a force balancing the disk thereby minimizing the pressure at which the disk will open.

3.8 Finish. Machined surfaces shall have a surface roughness average finish of 125 Roughness Average (Ra) or better. Contacting surfaces shall have a surface finish of 32 Ra. The finish symbols used in the specification are in accordance with ANSI/ASME Y14.36M.

3.9 Identification marking. Unless otherwise specified (see 6.2), valves shall be marked in accordance with MSS SP-25.

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3.10 Seat wrenches and grinding compound. When specified for valves with renewable seats (see 6.2), seat ring wrenches shall be furnished. The ratio of the number of wrenches to the number of valves shall be as specified (see 6.2). When specified for regrinding-type valves (see 6.2), regrinding compound shall be furnished.

3.11 Workmanship.

3.11.1 Fabrication. Metal used in the fabrication of equipment shall be free from fractures, splits, punctures, dents, deterioration, and shall not be bowed or malformed. The metal shall have no sharp edges, burrs, slivers, or splinters.

3.11.2 Welding. Welding shall be in accordance with S9074-AR-GIB-010/278 and welding procedures shall be in accordance with S9074-AQ-GIB-010/248. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.11.3 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided as necessary, and all bolts, nuts, and screws shall be tight.

3.11.4 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect that reduces the castings ability to perform its intended function. Non-destructive testing of castings shall be in accordance with 4.6.3.

3.12 Performance.

3.12.1 Shock. Valves shall withstand the shock requirements of grade A, hull mounted, class I, type C in accordance with MIL-DTL-901 and MIL-STD-798 (see 4.6.4).

3.12.2 Vibration. Valves shall withstand the type I vibration requirements in accordance with MIL-STD-167-1 (see 4.6.5).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of the examination of 4.5 and tests of 4.6. One valve of each type and class shall be subjected to first article inspection.

4.3 Conformance inspection. Conformance inspection shall be performed on samples selected in accordance with 4.4. This inspection shall include the visual and dimensional examination of 4.5 and the tests of 4.6.

4.4 Sampling.

4.4.1 Sampling for examination and tests. A random sample of valves shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at general inspection level II and the acceptance quality limit (AQL) specified (see 6.2).

4.4.2 Lot. A lot shall consist of all valves of the same classification produced under the same conditions and offered for delivery at one time, as specified (see 6.2). A sample valve shall consist of one completely assembled check valve.

4.5 Visual and dimensional examination. Sample valves selected shall be visually and dimensionally examined to verify compliance with the nonoperational requirements of this specification. Each of the sample valves selected shall be examined in accordance with table V and table VI. Any valve in the sample containing one or more defects shall be rejected, and if the number of defective valves in any sample exceeds the acceptable number for that sample, the lot represented by the sample shall be rejected. All cast pressure retaining components shall be visually inspected in accordance with S9074-AR-GIB-010/278.

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4.6 Tests. Tests shall be as specified in 4.6.1 and 4.6.2. Any leakage exceeding the maximum permissible rates shall be the cause of rejection of the valves unless corrective action is taken and the valves are satisfactorily retested. Unless a commercial laboratory is used for inspection, the manufacturer shall furnish all measuring and testing equipment, properly calibrated, required to conduct the examination and tests.

4.6.1 Pressure tests (bronze-body valves). Tests for bronze-body valves shall be in accordance with MSS SP-80.

4.6.2 Pressure tests (steel-body valves). Pressure testing for steel-body valves shall be in accordance with MSS SP-61.

4.6.3 Non-destructive testing (other than pressure testing). Cast-body valves shall require additional non-destructive testing (radiographic, magnetic particle, liquid penetrant) in accordance with S9074-AR-GIB-010/278, Category 3 when system design and service parameters require and shall meet the acceptance criteria specified therein. When specified (see 6.2), non-destructive testing of castings is required per system or design pressure requirements based on where the valve will be used in accordance with S9074-AR-GIB-010/278, Category 3.

4.6.4 Shock test. Each valve design shall be shock tested to determine conformance to 3.12.1. After shock testing, valves shall be tested in accordance with 4.6.1 or 4.6.2, as applicable.

4.6.5 Vibration test. Each valve design shall be tested to determine conformance to 3.12.2. After vibration testing, valves shall be tested in accordance with 4.6.1 or 4.6.2, as applicable

TABLE V. Examination and tests.

Examination/Test	Requirement	First Article	Conformance
Visual and dimensional examination	3.2 and 3.4-3.11	4.5	4.5
Pressure tests (bronze-body valves)	3.3 and table I	4.6.1	4.6.1
Pressure tests (steel-body valves)	3.3 and table II	4.6.2	4.6.2
Non-destructive testing (cast-body valves)	3.11.4 and table I and II	4.6.3	4.6.3
Shock testing	3.12.1	4.6.4	X
Vibration testing	3.12.2	4.6.5	X

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TABLE VI. Classification of defects.

Categories	Major Defects
101	Group, type, style, trim, end connection, size, class, or potable water use not as specified (see 1.2).
102	Material not as specified (see 3.2).
103	Valves of the same classification not physically and mechanically identical (see 3.4).
104	Valve body not as specified (see 3.6.1).
105	Caps or covers not as specified (see 3.6.2).
106	Seat not as specified (see 3.6.3).
107	Flow control device not as specified (see 3.6.4).
108	End connection not as specified (see 3.6.5).
109	Bosses, if required (see 6.2), not as specified (see 3.7.1).
110	Operating levers, if required (see 6.2), not as specified (see 3.7.2).
111	Finish roughness not as specified (see 3.8).
112	Identification markings not in accordance with MSS SP-25 (see 3.9).
113	Seat wrenches or regrinding compound, if required (see 6.2), not furnished (see 3.10).
114	Fabrication not as specified (see 3.11.1).
115	Welding not as specified (see 3.11.2).
116	Bolted connections not as specified (see 3.11.3).
117	Castings not as specified (see 3.11.4).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Check valves are intended for use in naval vessel and military piping systems to automatically prevent a reversal of flow through the line. Critical shipboard piping systems should be outfitted with regrindable or renewable check valves to permit at-sea and forward repair. Swing check valves offer less resistance to flow than lift checks and are generally used in systems where gate valves are employed. Lift checks are generally used where frequent reversals of direction are encountered and a pressure drop comparable to globe or angle valves is allowable. Ball lift checks are especially suited for viscous fluids. Non-metallic disks, when rated for media temperatures involved, result in quieter operation, will seat even when the media contains small particulate matter, and provide tighter closure at lower pressures, especially for air and gas service. Tilting disk check valves and spring-loaded check valves are characterized by rapid closure action, which tends to minimize flow reversal and resultant hydraulic hammer. Class 125, 150, and 200 bronze-body swing check valves and class 150 and 200 bronze-body horizontal, guided-disk lift check valves with metallic disks should be procured in accordance with MSS SP-80 if the regrindable seats covered by this specification are not required. Current stock of valves covered by this specification may be used until depleted; future requisitions for these valves should refer to the current

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version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Classification of valve required (see 1.2 and tables I, II, III, and IV.)

(1) Group.

(2) Type.

(3) Style (for lift check valves).

(4) Trim.

(5) End connection.

(6) Nominal size.

(7) Class.

(8) Potable water use.

c. When a sample valve is required for first article inspection and approval (see 3.1 and 4.2).

d. When stainless steel trim material is required for disk and seat rings on steel-body valves (see 3.2.5).

e. When optional, rather than standard, trim material is required for disks and seat rings (see table III).

f. When alternate facings for flanged valves are required (see table IV).

g. When spring-loaded disks are required on type IV, tilting disk check valves (see 3.3.3).

h. Type and rating of flanges between which wafer-body valves will be mounted (see 3.6.5).

i. When flange drilling in accordance with MIL-PRF-20042 is required (see table IV and 3.6.5).

j. Number and location of tapped bosses, if required, when tappings should be a size other than as required under MSS SP-45 (see 3.7.1).

k. When welded bosses are required in accordance with S9074-AR-GIB-010/278 (see 3.7.1).

l. When operating levers are required and, if so, whether levers should be weight- or spring-loaded (see 3.7.2).

m. When marking other than MSS SP-25 is required (see 3.9).

n. When special wrenches for removing seat rings are required and the quantity to be furnished; when regrinding compound is required and the quantity to be furnished (see 3.10).

o. Requirement for an AQL of 1.0 (see 4.4.1).

p. Lot size required (see 4.4.2).

q. Packaging requirements (see 5.1).

r. When non-destructive testing other than pressure testing is required (see 4.6.3).

6.3 First article. When first article inspection is required (see 6.2), the contracting officer should provide specific guidance to offerors whether the item should be a first article sample (see 3.1 and 4.2), and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternative bids unless specifically requested to do so in the solicitation.

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6.4 Subject term (key word) listing.

Bronze-Body
Butterfly Check Valve
Non-Metallic Disc
Regrinding
Renewable
Swing Check Valve
Tilting-Disk Check Valve

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 4820-2017-008)

Review activities:
Navy – MC, YD
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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